

5. You can upgrade your operating system only if you have a 32-bit processor running at 1 GHz or faster, at least 1 GB RAM, and 16 GB free hard disk space, or a 64-bit processor running at 2 GHz or faster, at least 2 GB RAM, and at least 32 GB free hard disk space. Express your answer in terms of u : "You can upgrade your operating system," b_{32} : "You have a 32-bit processor," b_{64} : "You have a 64-bit processor," g_1 : "Your processor runs at 1 GHz or faster," g_2 : "Your processor runs at 2 GHz or faster," r_1 : "Your processor has at least 1 GB RAM," r_2 : "Your processor has at least 2 GB RAM," h_{16} : "You have at least 16 GB free hard disk space," and h_{32} : "You have at least 32 GB free hard disk space."

Soln:
$$u \rightarrow ((b_{32} \wedge g_1 \wedge r_1 \wedge h_{16}) \vee (b_{64} \wedge g_2 \wedge r_2 \wedge h_{32}))$$

9. Are these system specifications consistent? "The system is in multiuser state if and only if it is operating normally. If the system is operating normally, the kernel is functioning. The kernel is not functioning or the system is in interrupt mode. If the system is not in multiuser state, then it is in interrupt mode. The system is not in interrupt mode."

Soln: p : The system is in multiuser state q : The system is operating normally r : The kernel is functioning s : The system is in interrupt mode.

$(p \leftrightarrow q)$ is True $(q \rightarrow r)$ is True $(\neg r \vee s)$ is True $(\neg p \rightarrow \neg s)$ is True

$\neg s$ is True.
By defn. of negation s is False. By defn. of OR, $\neg r$ is True. By defn. of negation r is False. By defn. of conditional, q is False. By defn. of conditional, $\neg p$ is False. By defn. of negation, p is True. By defn. of biconditional, $(p \leftrightarrow q)$ is False. (Contradiction)

$\therefore$ The specifications are not consistent